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EXPERIMENT 3

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Experiment

Design and simulate three separate network topologies in Cisco Packet Tracer — built around a Hub, a Switch, and a Router respectively — to compare how each device handles frame/packet forwarding.

Aim

To examine and compare the operation of a Hub, a Switch, and a Router by building three separate topologies in Cisco Packet Tracer and observing how data moves between the connected end devices in each case.

Objective

- To learn how a Hub, a Switch, and a Router each handle incoming traffic.
- To build and simulate small network topologies in Cisco Packet Tracer.
- To verify end-to-end connectivity between hosts using Ping/PDU tests.
- To compare broadcast-based forwarding (Hub) against learned/switched forwarding (Switch) and routed forwarding (Router).
- To understand where each device sits in terms of collision domains and broadcast domains.

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Software Requirements

Software

- Cisco Packet Tracer

Hardware (Virtual)

- Hub
- Switch
- Router
- Personal Computers (PCs)
- Copper Straight-Through Cables

Experiment Theory

A Hub is a Layer-1 device that simply repeats every incoming electrical signal out to all of its other ports, so all connected hosts share a single collision domain and a single broadcast domain. A Switch is a Layer-2 device that learns the MAC address on each port and forwards frames only to the port where the destination host actually sits, which removes unnecessary collisions while every host still remains in the same broadcast domain. A Router works at Layer-3, using IP addressing to move packets between separate networks, and each router interface forms its own broadcast domain — meaning routing is required for hosts on different subnets to reach each other.

Practical / Experiment Steps

Topology 1 – Hub-based Network

1. Drag one Hub component onto the workspace.
2. Connect three PCs to the Hub using Copper Straight-Through cables.
3. Assign a unique IP address to each PC on the same subnet.
4. Send a Ping / Simple PDU between the PCs.
5. Observe, in Simulation mode, that the Hub floods the frame out of every port.

Topology 2 – Switch-based Network

6. Drag one Switch component onto the workspace.
7. Connect four PCs to the Switch.
8. Assign IP addresses to all four PCs on the same subnet.
9. Test connectivity between the connected PCs.
10. Observe how the Switch learns MAC addresses and forwards frames only to the correct port.

Topology 3 – Router-based Network

11. Place one Router and two Switches on the workspace.
12. Connect each Switch to a separate interface on the Router.
13. Connect two PCs to each Switch (four PCs in total).
14. Configure IP addresses for all PCs and for both router interfaces, one subnet per switch.
15. Test communication across the two subnets through the Router.

16. Observe, in Simulation mode, how the Router examines the destination IP and routes the packet to the correct interface.

Procedure

1. Open Cisco Packet Tracer.
2. Build the Hub topology and confirm communication between all attached PCs.
3. Build the Switch topology and confirm communication between all attached PCs.
4. Build the Router topology, connecting two switches to the router with two PCs on each switch.
5. Assign IP addresses to every PC and router interface as required.
6. Use Ping / Simple PDU to confirm successful communication in each topology.
7. Record the packet flow and outputs observed in Simulation mode.

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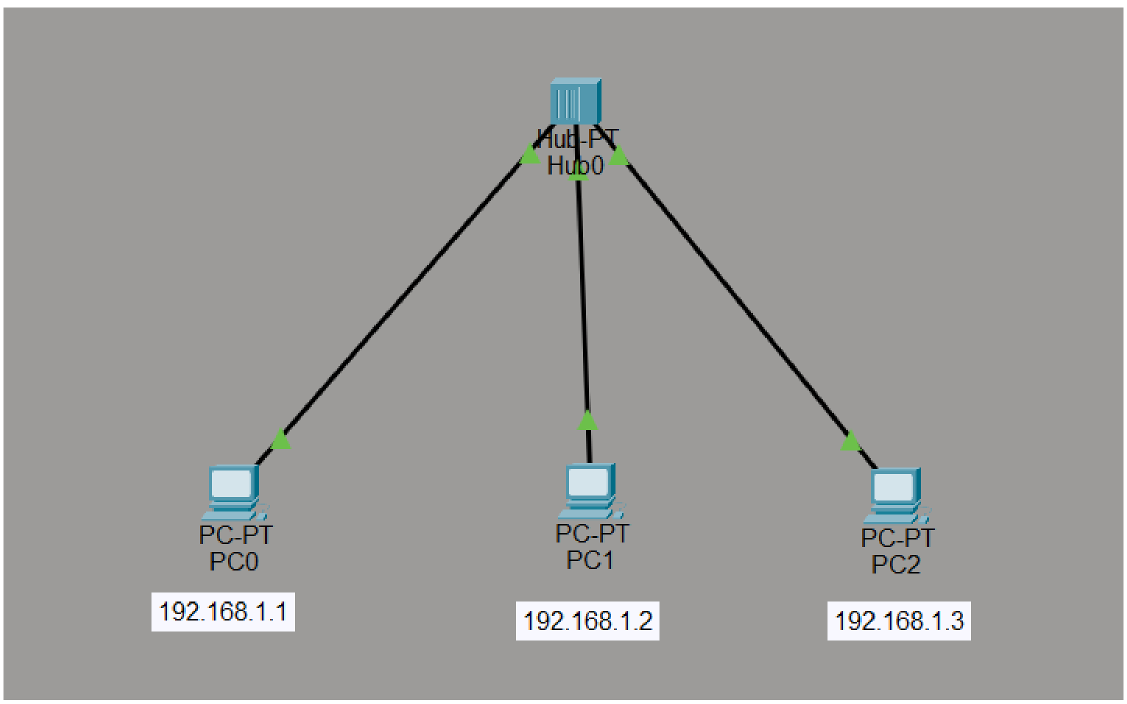
Input / Output Details

Devices Used

- 1 Hub
- 1 Switch
- 1 Router
- 2 Switches (for the Router topology)
- 11 PCs
- Copper Straight-Through Cables

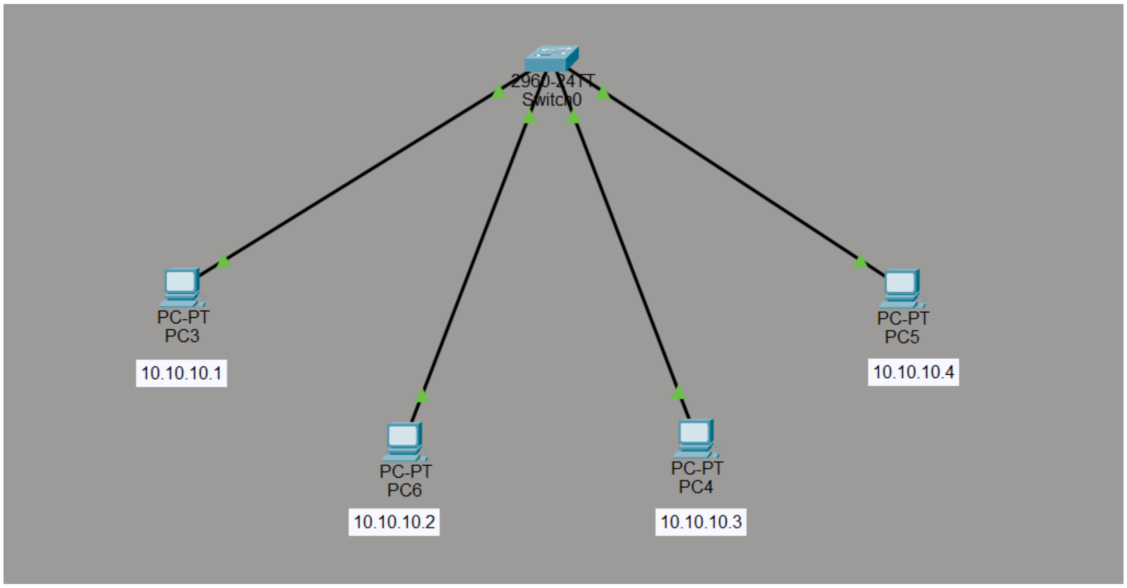
Step-wise Output

Step 1 – Hub Connected to Three PCs (192.168.1.0/24)



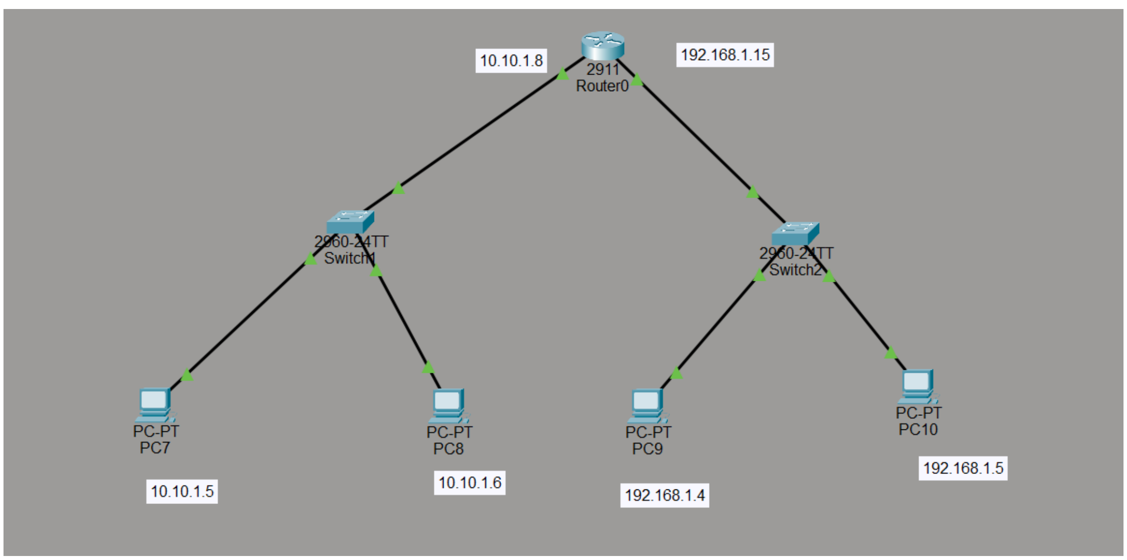
PC0, PC1, and PC2 are attached to Hub0 and assigned 192.168.1.1, 192.168.1.2, and 192.168.1.3 respectively. Because the Hub operates purely at Layer 1, a frame sent by any one PC is repeated out to every other port, regardless of the intended destination.

Step 2 – Switch Connected to Four PCs (10.10.10.0/24)



PC3, PC6, PC4, and PC5 are connected to Switch0 and configured with 10.10.10.1 through 10.10.10.4. Unlike the Hub, the Switch builds a MAC address table and forwards each frame only out of the port leading to its actual destination.

Step 3 – Router Connected to Two Switches, Two PCs per Switch



Router0 connects Switch1 (subnet 10.10.1.0/24, interface address 10.10.1.8) to PC7 (10.10.1.5) and PC8 (10.10.1.6), and connects Switch2 (subnet 192.168.1.0/24, interface address 192.168.1.15) to PC9 (192.168.1.4) and PC10 (192.168.1.5). Traffic between the two subnets must pass through the Router's routing table to reach its destination, unlike the flat Layer-2 forwarding seen with the Hub and Switch.

Learning Outcome

After performing this experiment, I was able to:

- Understand the working principle of a Hub, a Switch, and a Router.
- Differentiate between the three devices based on how each forwards traffic and the collision/broadcast domains they create.
- Design simple Hub, Switch, and Router-based topologies in Cisco Packet Tracer.

- Configure IP addressing and verify communication between connected devices.
- Analyze the role each networking device plays in data transmission and routing.